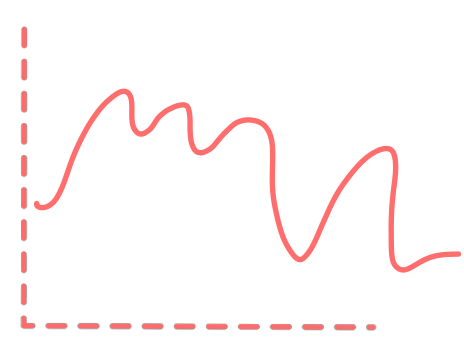


Types of Electrical Signals

Voltage is relative to Ground (GND)

Analog Signal
Continuous Value
Susceptible to noise



Digital Signals

Analogous to binary

Discrete Either 0 or 1

NOT 0V or 1V but rather HIGH or LOW Value

LOW is usually GND and HIGH is usually the component's max voltage

Usually follows strict formatting

Ways to read signals

Multimeters

- useful for reading a voltage without serious fluctuations
- Can use DC for constant values

Oscilloscope

Form of a Microcontroller

Mini Computer that goes on PCBs

Allows you to read/send electrical signals

Pins on Microcontrollers

Small metal rods sticking out the side of the chip

Pins connect to the traces on PCBs

Pins are the physical connection for where a microcontroller

ADC = Analog $\rightarrow$ digital

DAC = Digital $\rightarrow$ Analog

Can Bus Basics

Ways to communicate using digital signals over environments with significant disturbances

Multiple MCUs can tap into the same CAN bus

- All can read the same data being sent around the car

Can Waveforms

When a digital 0 is sent both wires remain at the same recessive voltage

Equals High and Low digital signals

When a digital 1 is sent both wires

CAN Data Packing

ID

Message ID unique to each type of message

DLC

Specifies the length of the data field

DATA

All of the binary being sent in the message up to 8 bytes

Arbitration

CAN Devices time synchronize to start transmission in predefined windows

CAN message ID's represent priorities

A system called arbitration allows for packet prioritization with no overhead

Firmware with MCUs

